

Locomotion and Movement

– The ability of an organism to move from one place to another or move its body parts.

1. INTRODUCTION

- **Movement** is a fundamental characteristic of **living organisms**.
- It helps in **survival**, **growth**, **interaction** and **reproduction**.
- The movement may involve the whole body or only a part of it.
- It is brought about by **muscular system** with the help of **skeletal system**.

2. TYPES OF MOVEMENT

(A) ON THE BASIS OF LOCATION

- (i) **Amoeboid Movement** – Movement by pseudopodia.

Example: Amoeba



- (ii) **Ciliary Movement** – Movement by cilia.

Example: Paramecium, Oviduct



- (iii) **Flagellar Movement** – Movement by flagella.

Example: Euglena, Human Sperm



(B) ON THE BASIS OF NATURE

- (i) **Non-directional Movement** – Random, no fixed direction.

Example: Brownian movement

- (ii) **Directional Movement** – Towards or away from a stimulus.

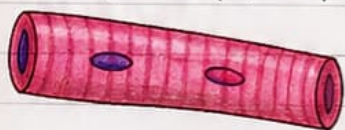
Example: Phototropism in plants

3. MOVEMENT IN HUMAN BEINGS

- Our body shows movement due to the action of muscles on bones.
- It is a coordinated effort of **muscular**, **skeletal** and **nervous** systems.
- Three types of muscles are present in the body.

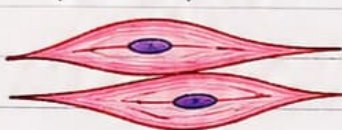
(i) Skeletal Muscle

- Striated (striped)
- Voluntary
- Attached to bones
- Long, cylindrical fibres
- Example: Biceps, Triceps



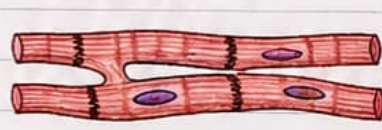
(ii) Smooth Muscle

- Non-striated
- Involuntary
- Present in walls of organs (e.g. intestines)
- Spindle shaped fibres



(iii) Cardiac Muscle

- Striated
- Involuntary
- Present only in heart
- Branched fibres with intercalated discs



– Did You Know? –

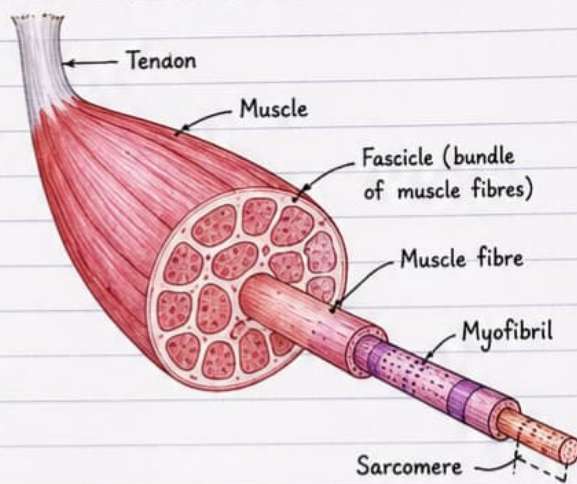
Over **600 muscles** work together in human body to perform thousands of movements every day!

4. MUSCULAR SYSTEM

- Muscles bring about movement by **contracting** and **relaxing**.
- In humans, muscles are of three types:

(i) Skeletal Muscle	(ii) Smooth Muscle	(iii) Cardiac Muscle
- Striated - Voluntary - Attached to bones - Brought about body movements - Example: Biceps	- Non-striated - Involuntary - Present in walls of internal organs - Example: Stomach - Intestine	- Striated - Involuntary - Present only in heart - Shows rhythmic contraction - Example: Heart

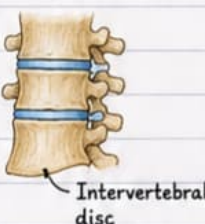
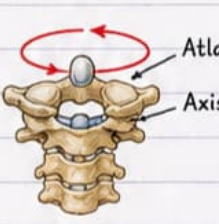
Structure of Skeletal Muscle



5. JOINTS

- Joints are the places where **two or more bones** meet.
- Joints give **flexibility** and help in movement.

Types of Joints

(i) Fibrous (Immovable)	(ii) Cartilaginous (Slightly movable)	(iii) Ball and Socket (Freely movable)	(iv) Hinge (Freely movable)	(v) Pivot (Freely movable)
No movement allowed.	Limited movement allowed.	Movement in all directions.	Movement in one plane only.	Movement of rotation.
				
Example: Joints in skull	Example: Joints between vertebrae	Example: Shoulder joint, Hip joint	Example: Elbow joint, Knee joint	Example: Joints between atlas and axis

6. MOVEMENT OF SKELETAL MUSCLES

- Muscles can only **pull**, not push.
- Muscles work in pairs of **antagonistic muscles**.
- When one muscle contracts, the other relaxes to bring about movement.

Antagonistic Pair in Human Arm (Example)

When Biceps Contracts:

- Biceps shortens
- Triceps relaxes
- Forearm bends at elbow (**flexion**)



When Triceps Contracts:

- Triceps shortens
- Biceps relaxes
- Forearm straightens at elbow (**extension**)



- Movements at joints are of **three** types:

- **Flexion** - Decreasing the angle between bones.
- **Extension** - Increasing the angle between bones.
- **Rotation** - Turning of a bone around its axis.

Other Movements

- ★ **Abduction** - Movement away from the midline of the body.
- ★ **Adduction** - Movement towards the midline of the body.
- ★ **Circumduction** - Circular movement.



Movement is Life, Life is Movement.



7. MOVEMENT IN PLANTS

- Plants do not have muscles, yet they show **movements**.
- These movements are usually in response to stimuli.
- They are of **two types**:

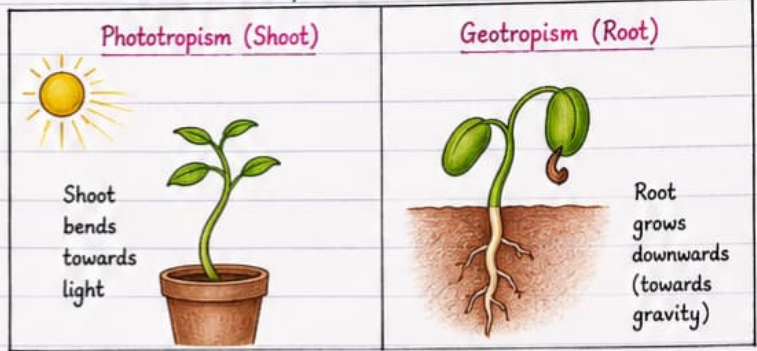
(i) **Growth dependent movements** – These are slow, irreversible and caused by growth.

Example: Phototropism, Geotropism in roots.

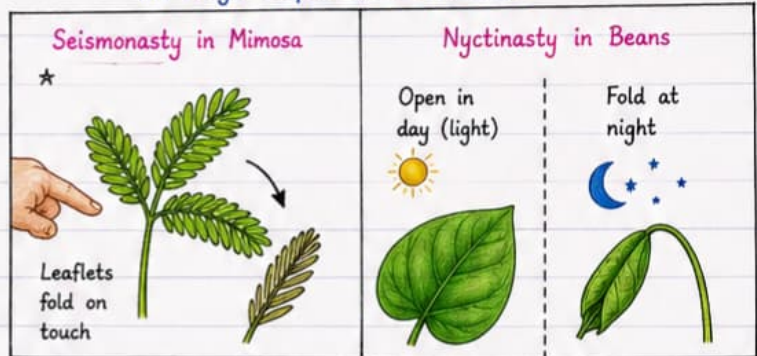
(ii) **Turgor dependent movements** – These are fast, reversible and caused by changes in turgor pressure.

Example: Seismonasty in Mimosa, Nyctinasty in leaves of beans.

(ii) Growth Dependent Movements



(ii) Turgor Dependent Movements



8. DISORDERS AND DEFICIENCIES OF MUSCULAR SYSTEM

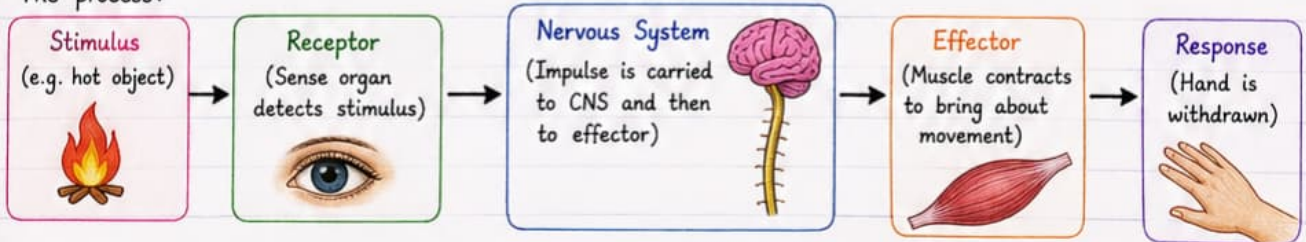
- Myasthenia Gravis** – Autoimmune disorder in which antibodies attack neuromuscular junction, causing muscle weakness.
- Muscular Dystrophy** – A group of genetic disorders causing progressive weakness and wasting of muscles.
- Tetany** – Involuntary muscle contractions due to low Ca^{2+} in blood.
- Arthritis** – Inflammation of joints causing pain, stiffness and swelling.

Did You Know?

Laughter, walking, blinking, chewing, even heartbeat – all are movements controlled by muscles!

9. COORDINATION OF MOVEMENT

- Movement is not a random event; it is **well coordinated**.
- It involves the **nervous system**, **muscular system** and **skeletal system**.
- The process:



10. IMPORTANCE OF MOVEMENT

- ★ Helps in maintaining posture and balance.
- ★ Enables us to perform various daily activities.
- ★ Helps in locomotion from one place to another.
- ★ Essential for circulation, digestion and respiration.
- ★ Helps in protection (e.g. withdrawal of hand from hot object).

Key Points to Remember

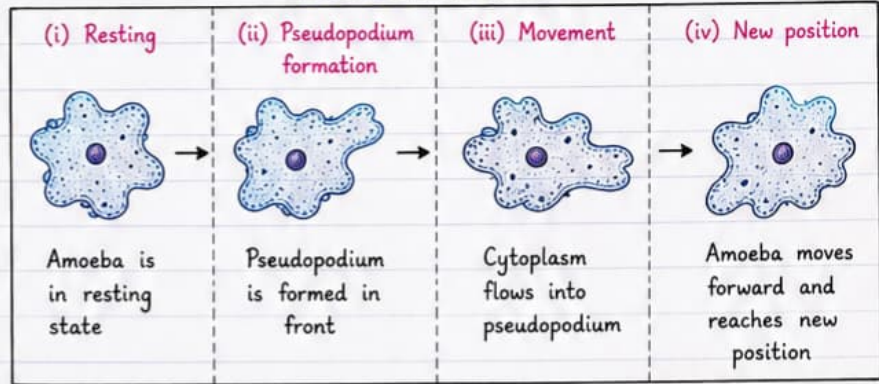
- ✓ Movement is life.
- ✓ Muscles + Bones + Joints work together.
- ✓ Nervous system coordinates all movements.
- ✓ Plants also show movements in response to stimuli.

Keep moving, keep growing!

11. MOVEMENT BY REARRANGEMENT OF CELL SHAPE

- Some organisms move by **changing** their **shape**.
- This type of movement **does not** involve muscles.
- It is seen in **Amoeba** and certain **WBCs**.
- Process:** Pseudopodia are formed in the direction of movement and the cell body flows into it.

Locomotion in Amoeba



12. MOVEMENT IN ANIMALS – AT A GLANCE

Type of Movement	Main Structure Involved	Example	Key Feature
Amoeboid	Pseudopodia	Amoeba	Irregular shape, slow movement.
Ciliary	Cilia	Paramecium	Numerous short cilia beat in coordination.
Flagellar	Flagellum	Euglena	One or few long flagella whip the water.
Muscular	Muscles	Human, Fish	Muscles contract and relax to bring about movement.
Rearrangement of Cell Shape	Pseudopodia	Amoeba, WBCs	Change in shape of cell helps in movement.

13. EXAMPLES FROM NATURE

- ★ **Earthworm** – Uses circular and longitudinal muscles for crawling.
- ★ **Snail** – Uses muscular foot for slow movement.
- ★ **Fish** – Uses body muscles and fins for swimming.
- ★ **Human** – Walks, runs, writes, speaks – all due to coordinated muscular and skeletal actions.



14. SUMMARY

- ✓ Movement is essential for **survival and growth**.
- ✓ It is brought about by the action of **muscles**.
- ✓ Muscles are attached to **bones** by **tendons**.
- ✓ Joints provide **flexibility** and different types of movement.
- ✓ Plants show movements in response to **stimuli** or due to **growth**.
- ✓ Coordination of **nervous**, **muscular** and **skeletal** systems is vital for movement.
- ✓ Movement is beautifully designed in both plants and animals.

Remember!

Movement = Life

Stillness = Stagnation

Keep moving, keep growing!

Quick Revision

- ★ Muscles → Contract & Relax
- ★ Bones → Support & Shape
- ★ Joints → Flexibility
- ★ Nerves → Control & Coordinate

